

TOOTH FAIRY: A DENTAL CLINIC APPOINTMENT MANAGER SYSTEM

**CPE21S4
Group 3**



TEAM 3 MEMBERS

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PROJECT OVERVIEW

PROBLEMS

- According to Gomez et al. (2023), accurate appointment capacity calculations and staffing allocation are critical to sustaining hospital management quality and are difficult to do without digital technology.
- Ho et al. (2024) noted that a large number of dental clinics continue to use paper-based records, which restricts cooperation between dental specialists and makes it difficult to obtain patient information quickly.
- Clinics frequently struggle to improve appointment processes to ensure accessibility and efficiency (Alahmad and Alnafea, 2023, as referenced in Sihombing, 2024).

OBJECTIVES

1. To develop a comprehensive system for storing and managing patient information
2. Provide a scheduling system that allows doctors and staff to manage appointments.
3. Ensure proper documentation and tracking of patient treatments and medical histories
4. Enable staff to quickly and accurately search for patient records and appointment details.
5. To develop an easy-to-use system interface that enables ease of navigation of patient records and appointment schedules.

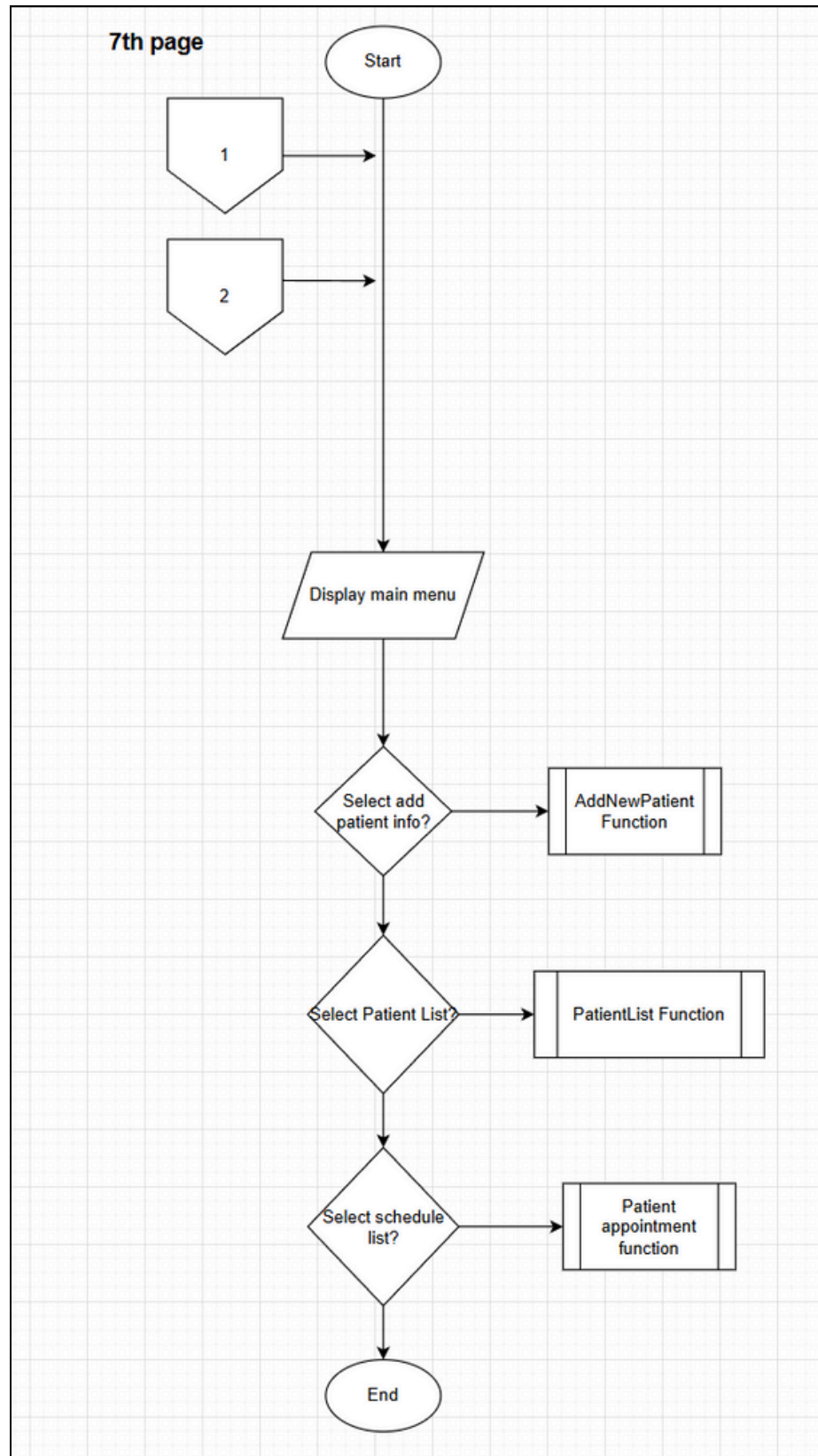
RATIONALE

- The project uses simple, standard C++ data structures: patients and appointments are stored in `std::vectors`, while `std::unordered_maps` handle ID tracking and doctor schedules.
- Helper types include `std::pair` for time ranges, `std::stringstream` for parsing, and lambdas for sorting and filtering.
- Core algorithms include time string parsing, unique ID generation, searching, deletion, appointment availability checks, and sorting by time.
- Input validation ensures correct data formats.
- The system prioritizes clarity and simplicity, with linear-time scans ($O(n)$ searches, $O(n \log n)$ sorts); efficiency could be improved with hash-based lookups or indexed scheduling maps for faster access.

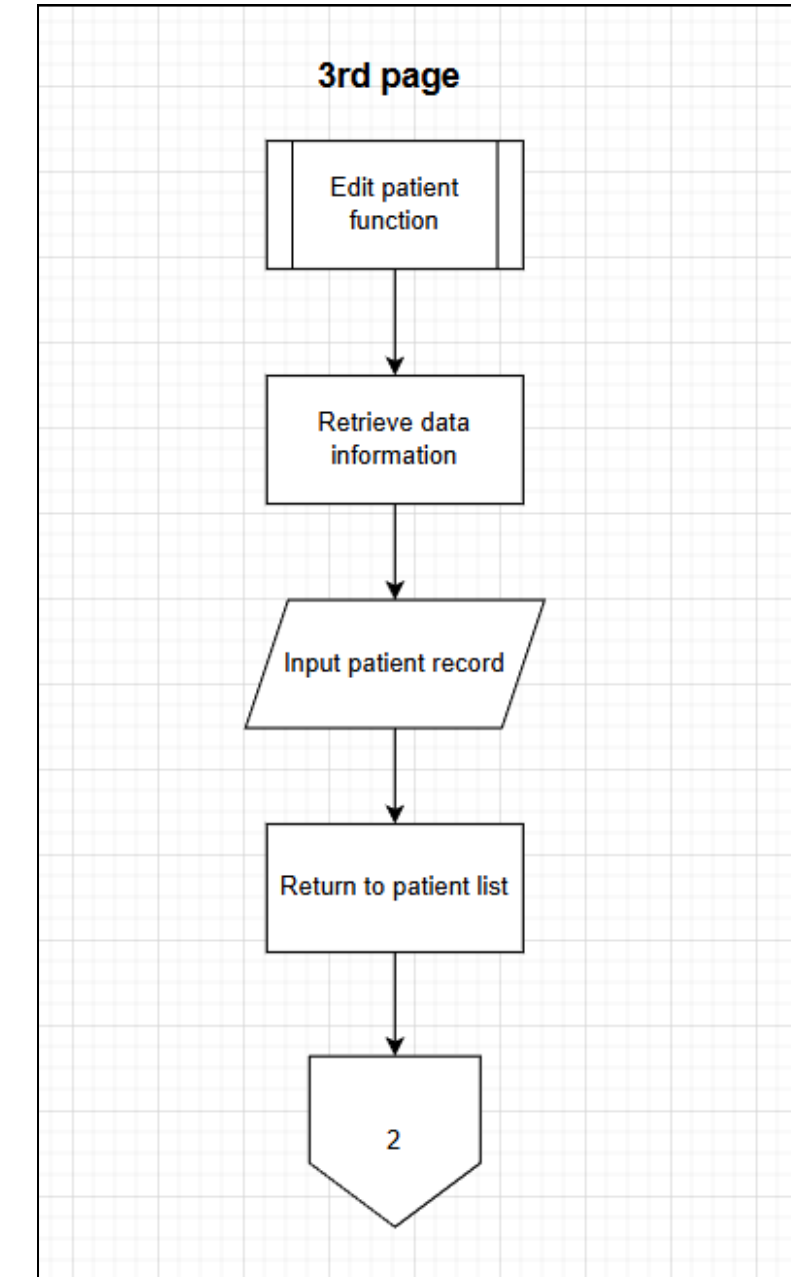
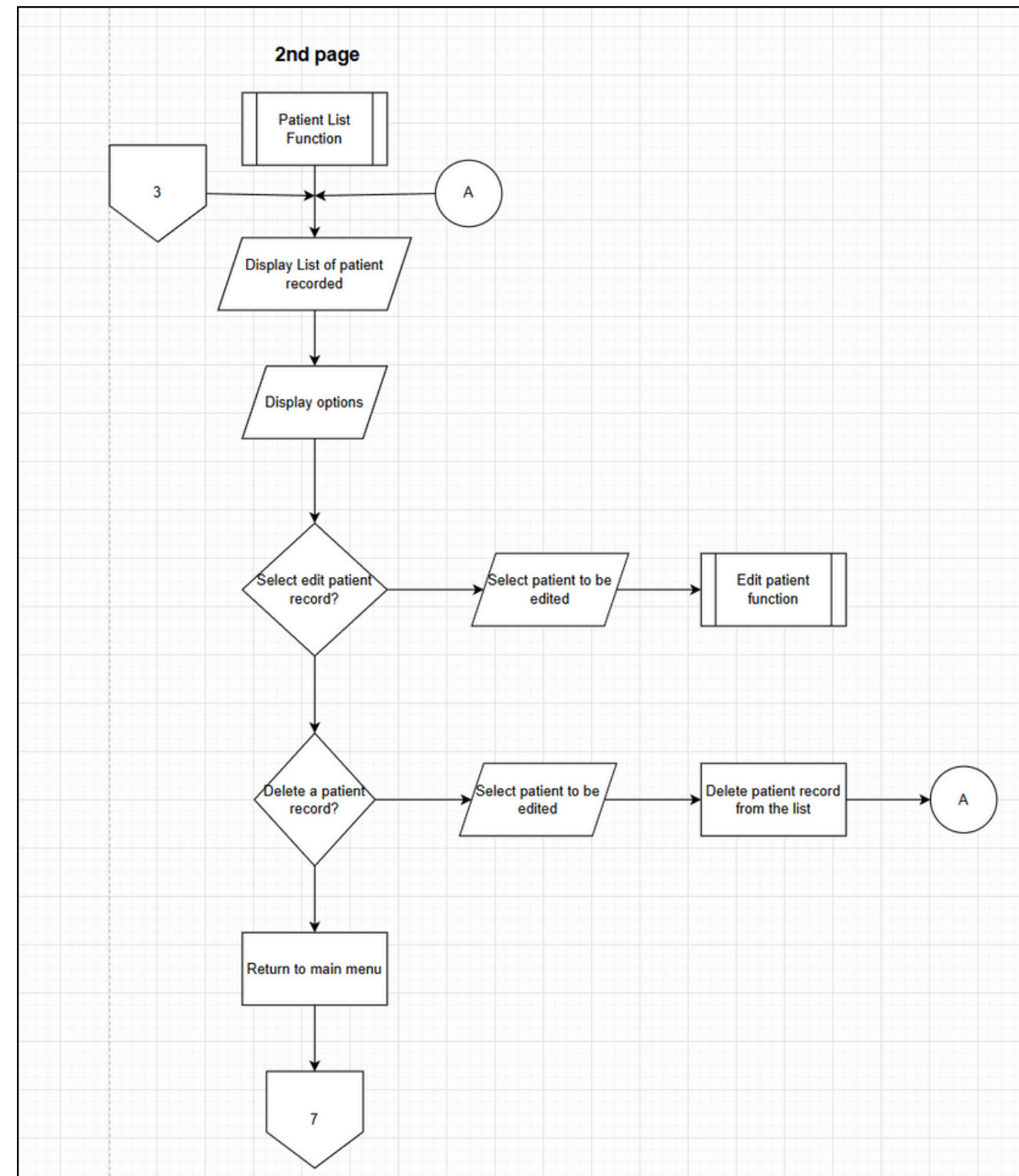
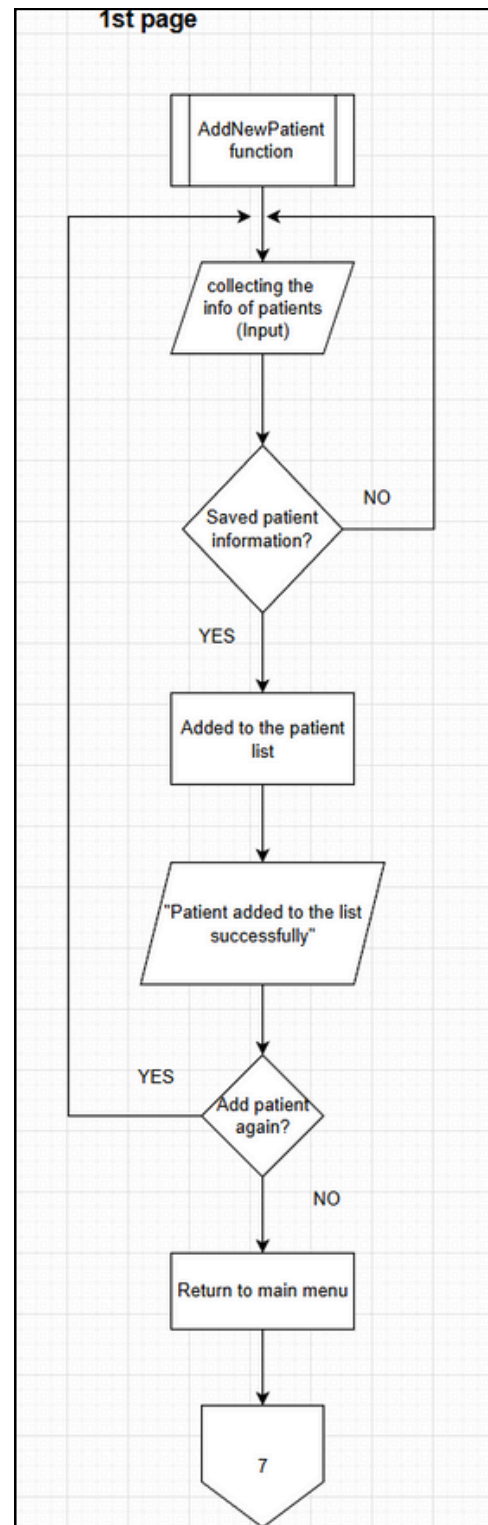
LIBRARIES

- `#include <vector>`
- `#include <iostream>`
- `#include <limits>`
- `#include <algorithm>`
- `#include <random>`
- `#include <unordered_map>`
- `#include <sstream>`
- `#include <cctype>`

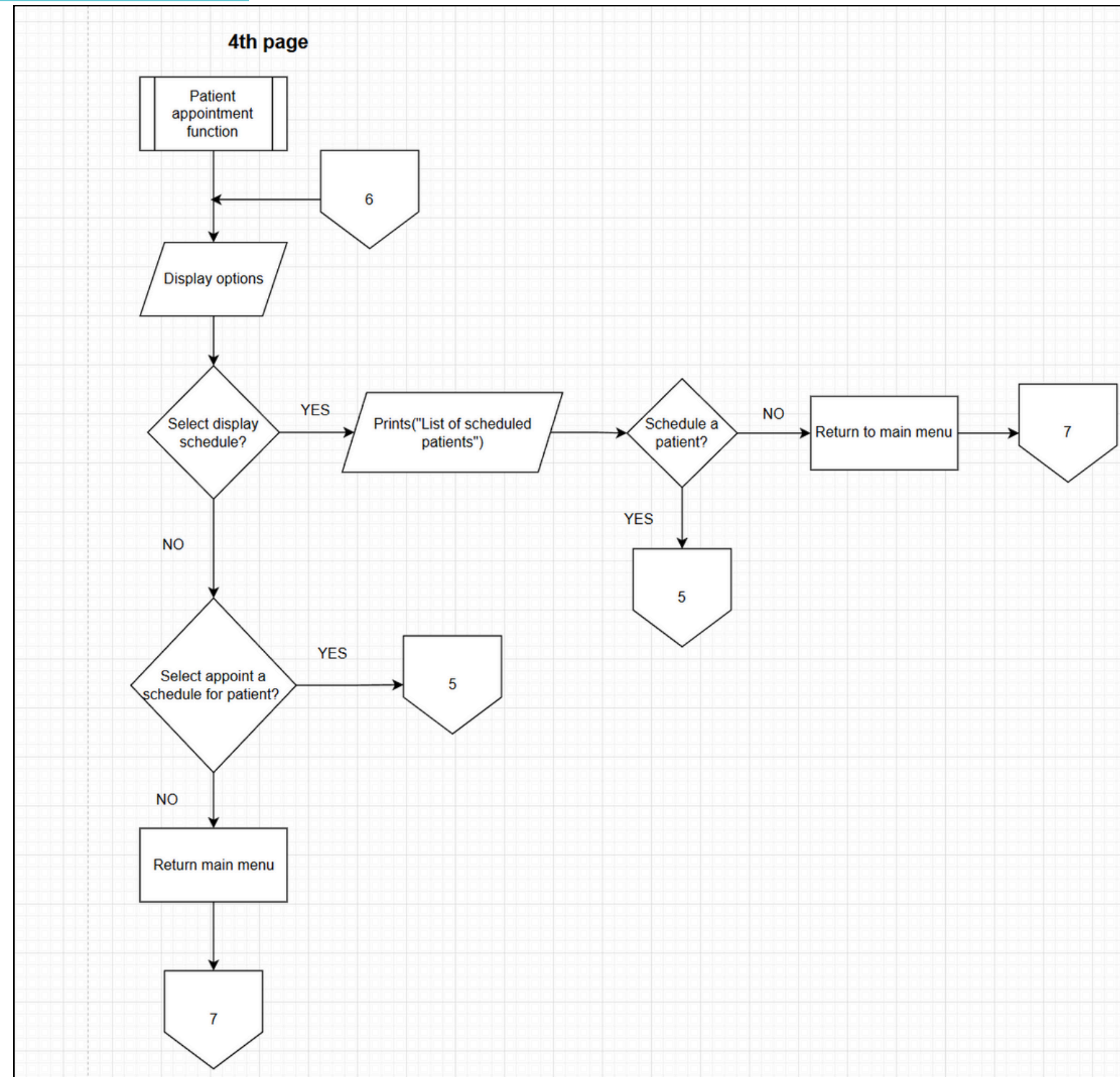
ALGORITHM FLOW



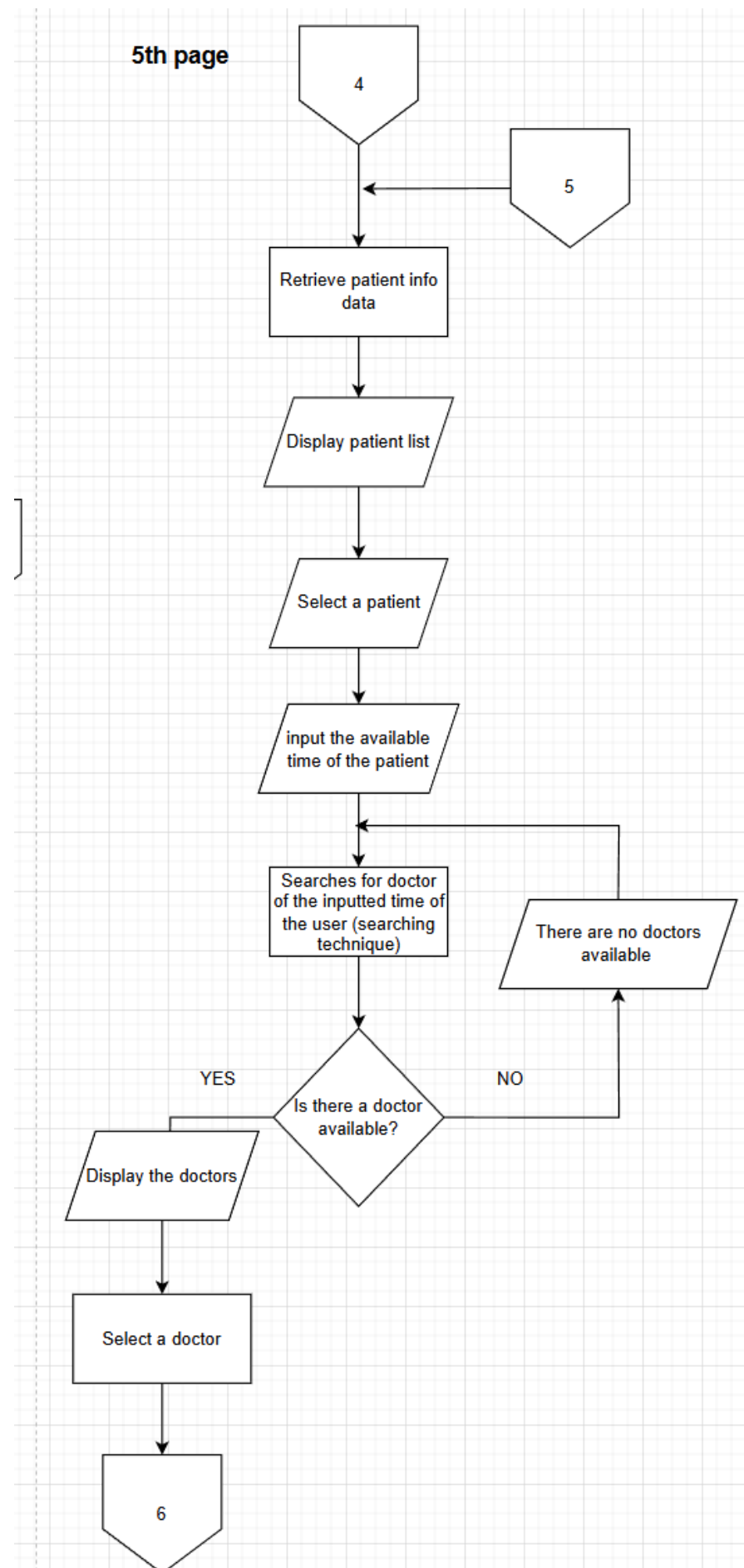
ALGORITHM FLOW



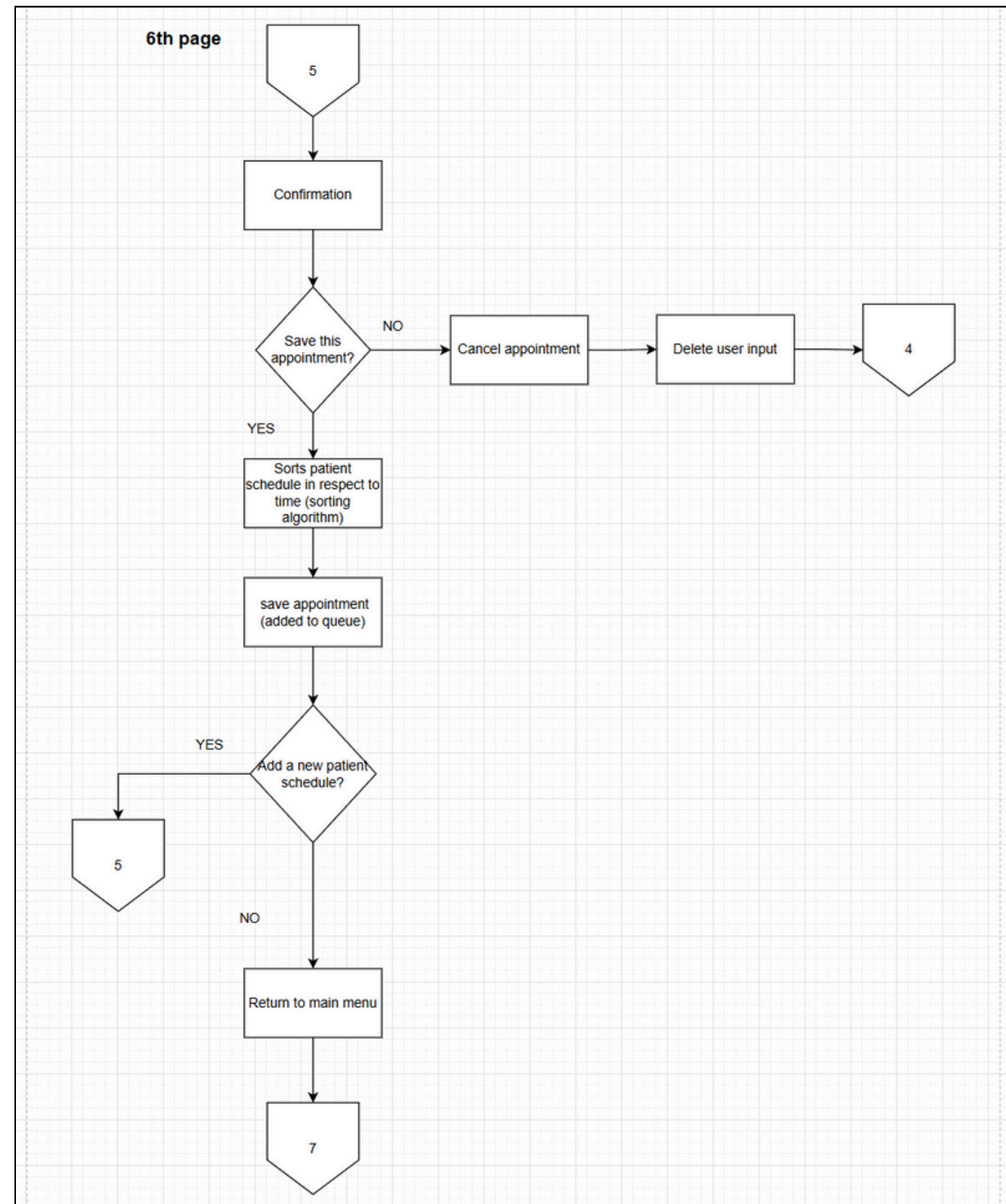
ALGORITHM FLOW



ALGORITHM FLOW



ALGORITHM FLOW



SAMPLE OUTPUTS

```
=====
Patient Management System
=====
1. Add New Patient Info
2. Patient List (Edit/Delete)
3. Patient Appointment Schedule
4. Exit
Select an option: 1
```

```
--- Patient List ---
ID: 4721, Name: Francis Nikko anody, Age: 20, Contact: 41234124123, Gender: m, History: No known medical history

Options:
1. Edit patient record
2. Delete a patient record
3. Return to main menu
Select an option: █
```

```
--- Patient List ---
ID: 4721, Name: Francis Nikko anody, Age: 20, Contact: 41234124123, Gender: m, History: No known medical history

Options:
1. Edit patient record
2. Delete a patient record
3. Return to main menu
Select an option: 2
Enter Patient ID to Delete: 4721
Patient ID 4721 deleted successfully.
```

```
--- Patient List ---
No patients recorded.
```

SAMPLE OUTPUTS

```
=====
Patient Management System
=====
1. Add New Patient Info
2. Patient List (Edit/Delete)
3. Patient Appointment Schedule
4. Exit
Select an option: 3
```

```
--- Patient Appointment Menu ---
1. Display schedule
2. Appoint a schedule for patient
3. Return to main menu
Select an option: 1
```

```
--- List of Scheduled Patients ---
No appointments scheduled.
Do you want to schedule a patient now? (Y/N): y
```

```
--- Schedule New Appointment ---
Available Patients:
ID: 9404, Name: Crishen Pulgago, Age: 21
Enter Patient ID for appointment (or enter 0 to select the most recently added patient): 9404
Enter preferred appointment time (e.g., 10:00 AM) or C to cancel: 10:00
```

```
Doctors available at 10:00:
1. Dr. Punay
2. Dr. Pizzaro
3. Dr. Crishen
4. Dr. Escalona
5. Dr. Owen
Enter doctor number to choose, or 0 to auto-assign the first available: 2
```

```
--- Confirmation ---
Patient: Crishen Pulgago (ID: 9404)
Time: 10:00
Doctor: Dr. Pizzaro
Save this appointment? (Y/N): y
Appointment saved successfully!
Do you want to schedule another appointment? (Y/N):
```

The background features a light gray field with abstract teal geometric elements. In the top-left and bottom-left corners, there are overlapping teal rectangles and diagonal lines. In the top-right and bottom-right corners, there are teal triangles and a grid of small teal circles. The text "THANK YOU" is centered in a large, bold, black sans-serif font.

**THANK
YOU**